Fiscal Year:	FY 2014	**Task Last Updated:**	FY 11/27/2013
PI Name:	Datta, Kamal M.D.		
Project Title:	High-LET Radiation, Reproductive Hormones and Chronic Inflammation: An Integrated Approach to Carcinogenesis Risk Estimate		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20057-0001	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
Start Date:	01/01/2013	**End Date:**	12/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2015, per NSSC information (Ed., 12/17/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AD58G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The overall goal of this project is to improve NASA risk estimates for space radiation-induced intestinal carcinogenesis. Inflammation and oxidative stress is intimately linked and both are strongly implicated in gastrointestinal (GI) carcinogenesis, and exposure to radiation results in production of reactive oxygen species (ROS) leading to oxidative stress. High levels of ROS are known to induce an apoptotic response but sustained sub-lethal oxidative stress has the potential to produce persistent modification of biomolecules including proteins, DNA, and lipids leading to alterations in their functions with adverse cellular consequences. Exposure of cells to stress is known to activate mitogenic signaling pathways such as EGFR/RAS/PI3KMAPK-related, which in turn stimulates proliferation and progression into mitosis. Persistent proliferative signals in the presence of DNA damage have been projected to propagate mutation in the progeny with survival advantage and carcinogenesis. Chronic sub-lethal oxidative stress could activate stress-sensitive kinases and transcription factors such as NFkB and AP1, and cause increased production of inflammatory cytokines. Although long-term effects of low linear energy transfer (low-LET) radiation such as gamma radiation on persistent oxidative stress and inflammation and how they relate to GI carcinogenesis have been investigated, information about the influence of high-LET cosmic radiation on GI cancer is scarce. Therefore, the overall goal of this proposal is to understand the effects of cosmic radiation on persistent oxidative stress and inflammation in the GI tract and how these alterations relate to GI carcinogenesis. In conjunction with our GI NSCOR program, the current proposal aims to compare the results of highly energetic charged (HZE) particle radiation to gamma radiation in mice to improve NASA's carcinogenesis risk estimates for GI cancer in male and female astronauts through a mechanistic understanding of the underlying biological processes. The specific aim for the proposal is therefore to characterize the roles of HZE radiation-induced persistent oxidative stress and inflammatory responses in gastrointestinal carcinogenesis.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	Gastrointestinal tumors are frequent in the U.S., and American Cancer Society estimates that there will be 142,820 new cases of colorectal cancer (CRC) with 50,830 persons predicted to die of the disease in 2013. Colorectal cancer when considered in men and women is the second leading cancer killer in the United States (http://www.cdc.gov/). Considering the high frequency of colorectal tumors in the American population, an even small increase by space radiation could have a major impact on risk estimates and planning of future manned space missions. Inflammation and cellular stress are critical elements of carcinogenesis and both are associated with alterations in biomolecules such as DNA. Because radiation is also known to initiate cellular oxidative stress and inflammatory responses, HZE radiation due to its qualitative difference from terrestrial radiation is expected to induce long-term stress and inflammatory responses thus increasing the risk of GI cancer.		
Task Progress:	Inflammation and oxidative stress is intimately linked and both are strongly implicated in gastrointestinal (GI) carcinogenesis. The overall aim is to characterize the effects of space radiation in triggering persistent stress and inflammatory responses and their potential effects on tumorigenesis in mouse intestine. This will involve quantitative and qualitative measurements of long-term cellular oxidative stress and related inflammatory responses in serum and in intestinal tissues after heavy ion and gamma-radiation. We have already carried out two beam runs (summer and fall, 2013) in the first year of this project at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory (BNL). We also seek to define how heavy ion radiation exposure affects the intestinal stem cell and its niche in relation to cellular stress and tumorigenesis. Because our GI tumor model, APC1638N, is in C57BL/6J background, we are performing our studies in C57BL/6J and APC1638N mice. During our first run (June 2013), we exposed C57BL/6J mice to 56Fe and gamma radiation. A group (7 mice per group) of 56Fe, gamma, and sham irradiated mice were sacrificed at 2 months after radiation and inflammatory response and oxidative stress studies are ongoing in the intestinal tissues. Remaining mice (7 mice per group) will be sacrificed at 12 months after exposure. Results at 2 months will be correlated to 12 month results for persistent effects. Inflammatory response analysis of the intestinal samples from 2 months post-exposure mice were performed and significant increase in pro-inflammatory markers were observed. During the first run we also exposed C57BL/6J mice to high (1 Gy/min) and low (0.01 Gy/min) rate 28Si radiation for a total dose of 1 Gy. Oxidative stress is being compared in colonic epithelial cells 2 months after exposure. During our second run (November 2013), we exposed APC1638N mice to 56Fe radiation. For a comparative study, we also exposed APC1638N mice to gamma radiation. Currently, these mice are being monitored for intestinal tumorigenesis and will be sacrificed at 150 days post-radiation. Additional beam time has been obtained for spring of 2014 and beam time has also been requested for summer and fall of 2014.		

Bibliography Type:	Description: (Last Updated: 06/25/2025)
Abstracts for Journals and Proceedings	Suman S, Fornace AJ Jr, Datta K. "Exposure to 56Fe radiation caused persistently increased serum levels of known colorectal cancer biomarkers in C57BL/6J mice." 59th Annual Meeting of the Radiation Research Society, New Orleans, LA, September 14-18, 2013. 59th Annual Meeting of the Radiation Research Society, New Orleans, LA, September 14-18, 2013. Presentation Number: PS3-28. Abstract available at: http://www.abstractsonline.com/Plan/ViewAbstract.aspx?sKey=c2100182-c052-4560-b7d4-ba46257d214c&cKey=6fb2f11d-fa32-401e-8ad4-dc67fb3189b8&mKey=01c994ce-9545-4cc7-8580-d74360fa8373 ; accessed 12/2/2013. , Sep-2013
Abstracts for Journals and Proceedings	Cheema AK, Suman S, Kaur P, Shay JW, Fornace AJ Jr, Datta K. "Distinct metabolomics profile and increased inflammatory markers in mouse intestine two months after heavy ion radiation exposure." 59th Annual Meeting of the Radiation Research Society, New Orleans, LA, September 14-18, 2013. 59th Annual Meeting of the Radiation Research Society, New Orleans, LA, September 14-18, 2013. Presentation Number: PS3-11. Abstract available at: http://www.abstractsonline.com/Plan/ViewAbstract.aspx?sKey=c2100182-c052-4560-b7d4-ba46257d214c&cKey=c60f9017-ebda-4d68-a340-ec6930646d8f&mKey=01c994ce-9545-4cc7-8580-d74360fa8373 ; accessed 12/2/2013. , Sep-2013